GROUP WORK PROJECT # 1 MScFE 660: RISK MANAGEMENT

Group Number: 18611

FULL LEGAL NAME	LOCATION (COUNTRY)	EMAIL ADDRESS	MARK X FOR ANY NON-CONTRIBUTING MEMBER
Virginia Hlatshwayo	South Africa	hlatshwayo.virginia@gmail.com	
Shaoheng Ding	China	dingshaoheng@msn.com	
Franck Delma Deba Wandji	France	debafranck@gmail.com	

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).

Team member 1	Virginia Hlatshwayo
Team member 2	Shaoheng Ding
Team member 3	Franck Delma Deba Wandji

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed.
Note: You may be required to provide proof of your outreach to non-contributing members upon request.

N/A

1 Problem Formulation

1.1 The Problem

Crude oil is simultaneously a physical commodity, a financial asset, and a geopolitical bargaining chip, and its price reflects the interaction of all three. A single, static statistical model that assumes one fixed relationship between oil prices and a handful of macroeconomic drivers struggles to cope with this environment: the relevant set of drivers and the strength of their relationships both change over time as the market moves between demand-driven, supply-driven, and purely speculative/financial regimes (Kilian; Hamilton). A risk manager who prices energy exposure, hedges an oil-linked book, or sets capital against commodity risk therefore faces two compounding problems: (i) identifying which set of financial, macroeconomic, and physical-market variables is actually driving the price at a given point in time, out of a large and only loosely related candidate set, and (ii) doing so in a way that is transparent enough to audit.

This project follows the framing used Alvi: treat crude oil price forecasting as a problem of structure discovery over a mixed panel of financial, macroeconomic, and geopolitical/physical-market series, rather than a problem of fitting one fixed regression equation. Given m candidate variables $X_1, \ldots, X_m$ (our data dictionary in Table 1 lists 24: 5 financial, 8 macroeconomic, and 11 geopolitical/physical-market series sourced from Yahoo Finance, FRED, and the EIA Short-Term Energy Outlook respectively), the task is to learn both which variables are conditionally dependent on which others, and the strength and direction of those dependencies, directly from the data.

1.2 Why Probabilistic Graphical Models Fit

A Bayesian (belief) network represents the joint distribution of $X_1, \ldots, X_m$ as a directed acyclic graph (DAG) in which each node's distribution is conditioned only on its direct parents, factorising the joint density as

$$p(x_1, \ldots, x_m) = \prod_{v=1}^{m} p\big(x_v \mid \mathrm{pa}(x_v)\big).$$

This representation is a natural fit for the oil-pricing problem for three reasons.

1. **It encodes conditional independence, not just correlation.** Many of our 24 candidate series are themselves highly correlated with each other (e.g. WTI and Brent futures, or the several EIA supply/demand series derived from the same underlying balance identity — see the correlation heatmap in Figure 4). A naive linear model regressing price on all 24 variables would be dominated by multicollinearity. A belief network instead asks which variables remain informative about the price once its other direct causes are already known, which is exactly the conditional-independence structure a DAG is built to represent.

2. **Structure learning requires little prior expert knowledge.** Because the graph itself (not just its parameters) is learned from data, the model does not require the analyst to pre-specify which of the 24 series matter, or how; it only requires selecting a reasonable candidate set and a learning algorithm (Section 6). This is valuable in a market where the dominant drivers of price are known to shift across regimes (demand shocks vs. supply shocks vs. pure financial sentiment), since a hand-specified structure risks encoding yesterday's regime.

3. **Inference and interpretation are transparent.** Once learned, a belief network supports exact probabilistic queries (e.g. "what is the probability that WTI falls given that OPEC spare capacity has risen and the US dollar has strengthened?") via standard inference algorithms, and every edge in the graph is independently inspectable and explainable to a risk committee — unlike the weights of an opaque machine-learning model.

1.3 Advantages Over Alternative Approaches

Relative to the two most natural alternatives for this problem, probabilistic graphical models offer specific, concrete advantages for our use case:

- **Compared to a single multivariate linear regression / VAR.** A linear model assumes one fixed functional form and one fixed set of "significant" regressors for the entire sample. Our stylized-facts analysis (Section 4) shows that oil returns are strongly non-Normal (excess kurtosis of 18.7 for WTI) and that cross-variable relationships are themselves unstable over time (the 63-day rolling correlation between WTI and the US dollar index swings repeatedly between -1 and $+1$ across the sample, Figure 5) — a single fixed linear structure cannot represent a relationship whose sign itself is time-varying, whereas a belief network's structure can be re-learned regime by regime.

- **Compared to an unconstrained machine-learning model (e.g. gradient boosting, a neural network).** Such models can achieve strong in-sample fit but provide little insight into why a forecast changed, which is unacceptable for risk-management sign-off. A belief network's DAG is a directly interpretable artifact: every arc corresponds to a specific, testable conditional dependence, and the Markov blanket of the price node (Section 7) tells the risk manager, in one step, the minimal set of variables that need to be monitored to screen the price node off from everything else in the system.

- **Combinability with regime-switching dynamics.** Because a belief network only encodes a static joint distribution, it is naturally paired with a Hidden Markov Model that first discretises each time series into a small number of latent regimes (bull/bear/stagnant), and then learns one belief network over the discretised states.

In summary, the core modelling choice of this project — a belief network over a mixed financial/macroeconomic/geopolitical panel, informed by a preceding regime-detection step — is chosen specifically because it matches the structural features we observe in the data (Section 4): non-Normal, regime-dependent returns; an unstable and large set of candidate drivers; and a need for an auditable, rather than opaque, decision process.

2 Data

2.1 Data Sourcing

Following Alvi's design (§3.1 of (Alvi)), the candidate variable set is split into three groups:

- **Financial (5 series).** Daily closes from Yahoo Finance: WTI and Brent continuous front-month futures, the ICE US Dollar Index, the S&P 500, and the 10-year US Treasury yield.
- **Macroeconomic (8 series).** Monthly series from the Federal Reserve Economic Data (FRED) API, matching Alvi's Table 3.2 exactly: energy CPI, oil & gas extraction capacity and capacity utilisation, three industrial-production indices (oil & gas extraction, crude petroleum & natural gas, and oil & gas support activities), total industrial production, and the oil & gas extraction producer price index.
- **Geopolitical / physical-market (11 series).** Monthly Short-Term Energy Outlook (STEO) series from the EIA Open Data v2 API, matching 11 of the 14 series in Alvi's Table 3.1 (OPEC/non-OPEC supply, OECD/non-OECD demand, US production and refinery input, OPEC spare/production capacity, and OECD/non-OECD stock changes). The remaining 3 of Alvi's 14 series (`STEO.RGDPQ_NONOECD.M`, `STEO.RGDPQ_OECD.M`, `STEO.FOREX_WORLD.M`) return no data against the current STEO vintage and have been retired or renamed since 2018; rather than guess a replacement series ID, we omit them and document the omission here explicitly.

One data-quality nuance specific to the EIA STEO series is worth flagging explicitly: each pull includes roughly a year of the EIA's own forward-looking forecast beyond the present day, which was truncated before any exploratory or stylised-facts analysis so that only realised history is analysed. The resulting data dictionary is given in Table 1.

Table 1: GWP1 data dictionary: 24 series across the financial, macroeconomic, and geopolitical/physical-market groups.

Name	Category	Source	Freq.	Units	Start	End
brent_futures	financial	Yahoo Finance	daily	n/a	2007-07-30	2026-10-06
sp500	financial	Yahoo Finance	daily	n/a	1990-01-02	2026-10-05
usd_index	financial	Yahoo Finance	daily	n/a	1990-01-01	2026-10-06

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Table 1: GWP1 data dictionary: 24 series across the financial, macroeconomic, and geopolitical/physical-market groups.

Name	Category	Source	Freq.	Units	Start	End
ust10y_yield	financial	Yahoo Finance	daily	n/a	1990-01-02	2026-10-05
wti_futures	financial	Yahoo Finance	daily	n/a	2000-08-23	2026-10-06
demand_non_oecd	geopolitical	U.S. Energy Information Administration (EIA)	monthly	n/a	1990-01-01	2026-10-01
demand_oecd	geopolitical	U.S. Energy Information Administration (EIA)	monthly	n/a	1990-01-01	2026-10-01
non_oecd_stock_change	geopolitical	U.S. Energy Information Administration (EIA)	monthly	n/a	2003-02-01	2026-10-01
oecd_inventory_balance	geopolitical	U.S. Energy Information Administration (EIA)	monthly	n/a	2003-01-01	2026-10-01
oecd_stock_change	geopolitical	U.S. Energy Information Administration (EIA)	monthly	n/a	2003-02-01	2026-10-01
opec_production_capacity	geopolitical	U.S. Energy Information Administration (EIA)	monthly	n/a	1990-01-01	2026-10-01
opec_spare_capacity	geopolitical	U.S. Energy Information Administration (EIA)	monthly	n/a	1993-01-01	2026-10-01
supply_non_opec	geopolitical	U.S. Energy Information Administration (EIA)	monthly	n/a	1993-01-01	2026-10-01
supply_opec	geopolitical	U.S. Energy Information Administration (EIA)	monthly	n/a	1993-01-01	2026-10-01
us_crude_production	geopolitical	U.S. Energy Information Administration (EIA)	monthly	n/a	1990-01-01	2026-10-01
us_refinery_input	geopolitical	U.S. Energy Information Administration (EIA)	monthly	n/a	1990-01-01	2026-10-01
capacity_oil_gas_extraction	macroeconomic	FRED (Federal Reserve Economic Data)	monthly	n/a	1972-01-01	2026-08-01
capacity_util_oil_gas_extraction	macroeconomic	FRED (Federal Reserve Economic Data)	monthly	n/a	1972-01-01	2026-08-01
cpi_energy	macroeconomic	FRED (Federal Reserve Economic Data)	monthly	n/a	1957-01-01	2026-08-01

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Table 1: GWP1 data dictionary: 24 series across the financial, macroeconomic, and geopolitical/physical-market groups.

Name	Category	Source	Freq.	Units	Start	End
indprod_crude_petroleum_ng	macroeconomic	FRED (Federal Reserve Economic Data)	monthly	n/a	1972-01-01	2021-04-01
indprod_oil_gas_extraction	macroeconomic	FRED (Federal Reserve Economic Data)	monthly	n/a	1972-01-01	2026-08-01
indprod_oil_gas_support	macroeconomic	FRED (Federal Reserve Economic Data)	monthly	n/a	1972-01-01	2026-08-01
indprod_total	macroeconomic	FRED (Federal Reserve Economic Data)	monthly	n/a	1919-01-01	2026-08-01
ppi_oil_gas_extraction	macroeconomic	FRED (Federal Reserve Economic Data)	monthly	n/a	1985-12-01	2026-08-01

2.2 Data Cleaning

Every series is passed through the same three-stage:

1. **Bad data.** Non-monotonic or duplicated timestamps, and economically impossible non-positive values, are treated as errors and set to missing. One deliberate exception is built in: the OECD and non-OECD inventory *stock-change* series are genuinely *signed* flow variables (a negative value denotes a net stock *build*, not an error), so the non-positive check is disabled for those two series only.

2. **Extreme outliers.** A robust, median-absolute-deviation (MAD) based modified z-score flags (but does not remove) observations with $|z| > 3.5$, so genuine price shocks remain visible in the sterilised series while still being documented.

3. **Missing values.** Slow-moving macroeconomic/geopolitical series are imputed by linear interpolation (appropriate given their short, infrequent gaps); the daily financial series instead use a volatility-matched Brownian-bridge simulation between the surrounding observed values, which avoids injecting artificial flatness into a price series across a longer gap (e.g. a data-vendor outage spanning several trading days).

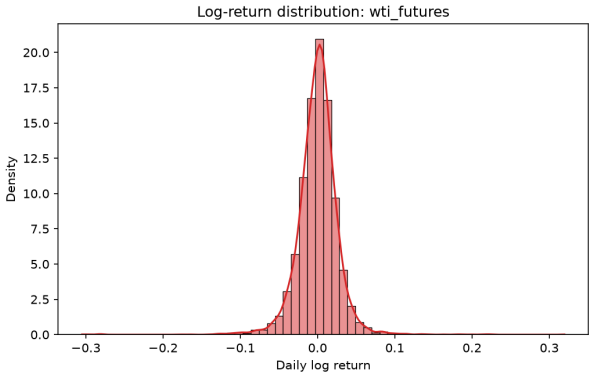
One specific finding is worth flagging on its own merits rather than leaving implicit in the narrative above: the raw WTI futures feed contains exactly one non-positive observation, corresponding to the historic 20 April 2020 episode when the expiring WTI contract briefly traded at

a negative price as storage capacity was exhausted and long holders could not take delivery. Our general-purpose bad-data rule (non-positive price is an error) is a defensible default for a panel of 24 heterogeneous series, but it has the side effect of treating this genuine, famous market event identically to a data error and imputing it away. We flag this transparently here, and again in Section 4, rather than silently losing the observation.

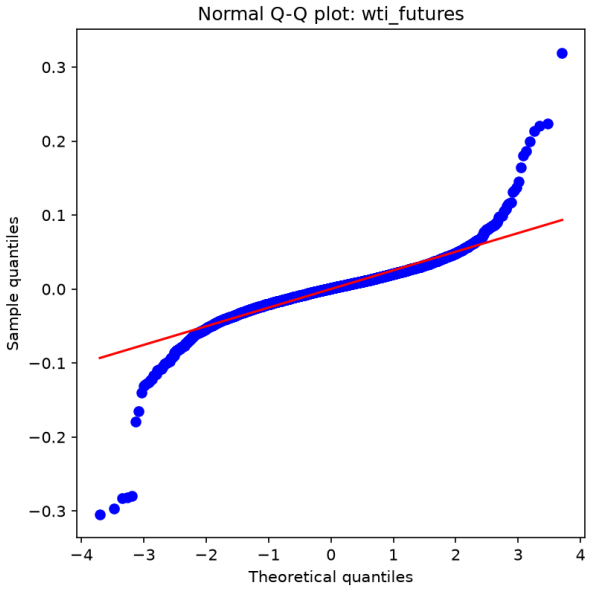
3 Exploratory Data Analysis

In this section we focus on the two benchmarks WTI and Brent, since "returns" are only a meaningful concept for a traded price series, not for a level/flow series such as OPEC spare capacity.

3.1 Distributional Plots



(a) WTI daily log-return distribution



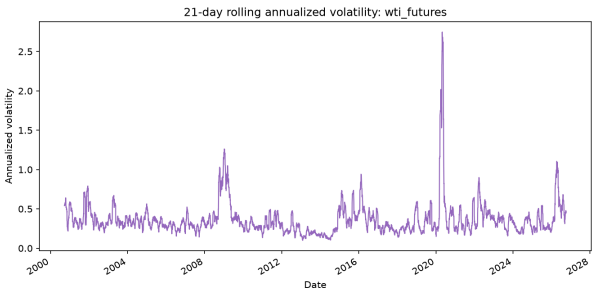
(b) WTI daily log-return Q-Q plot

Figure 1: Distributional diagnostics for WTI daily log returns.

3.2 Time Series Plots



(a) WTI price level



(b) 21-day rolling annualised volatility

Figure 2: WTI price level and rolling volatility, full sample.

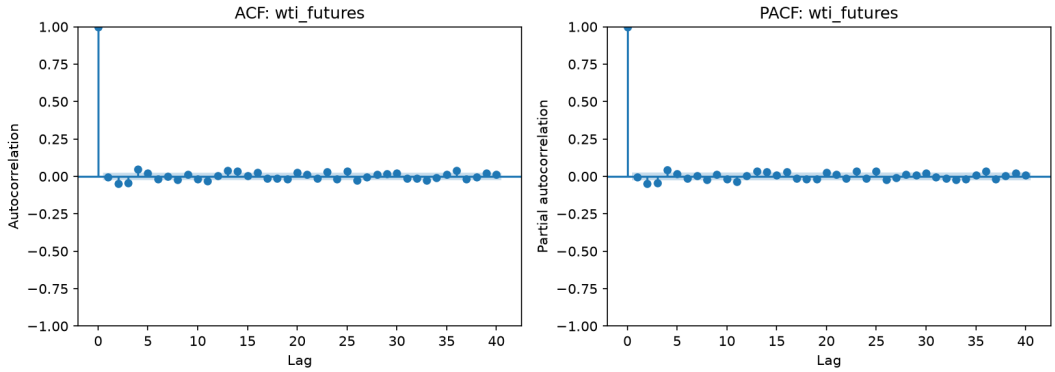


Figure 3: ACF and PACF of WTI daily log returns (40 lags).

3.3 Multivariate Plots

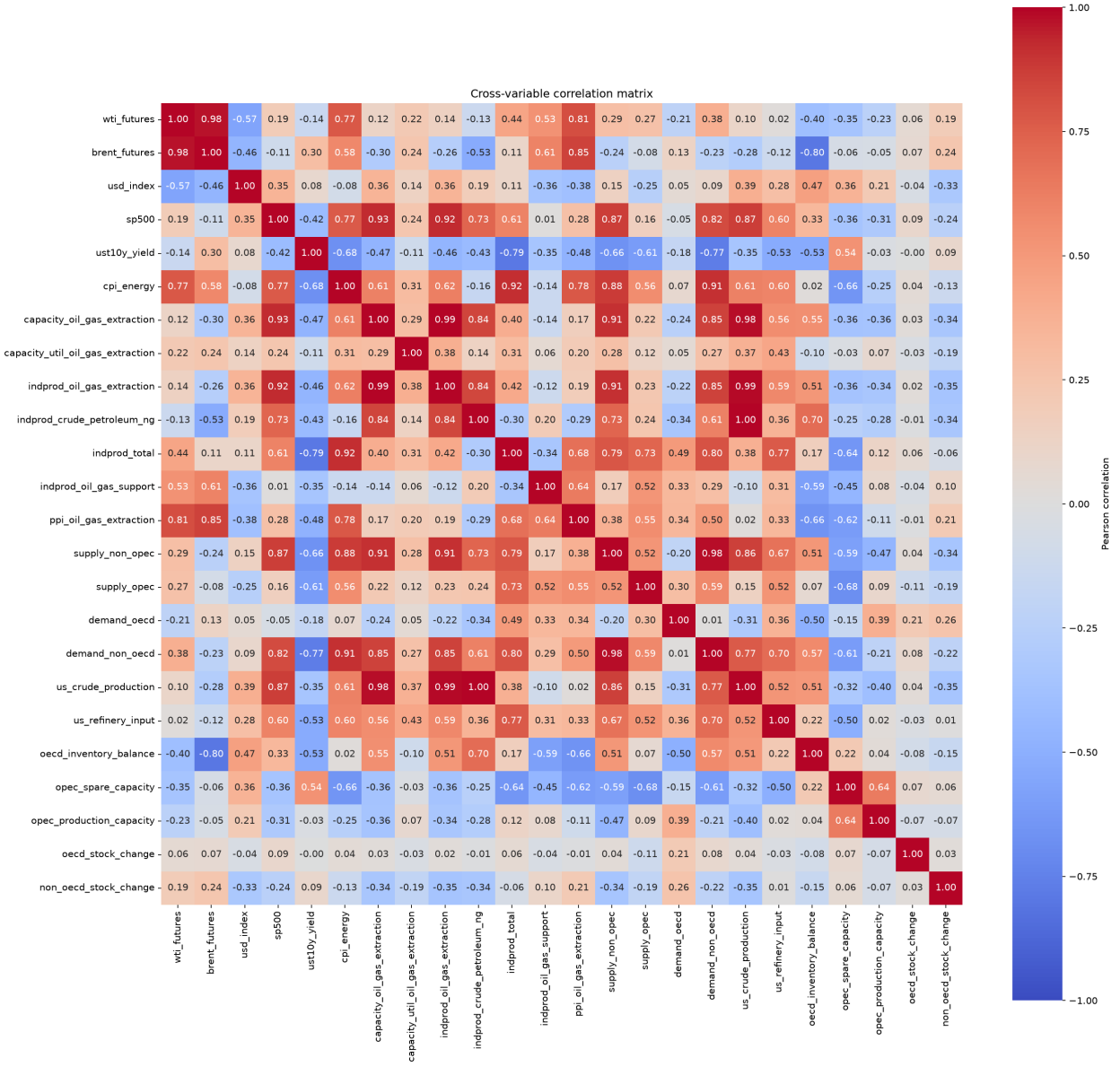


Figure 4: Pearson correlation matrix across all 24 monthly-aligned series.

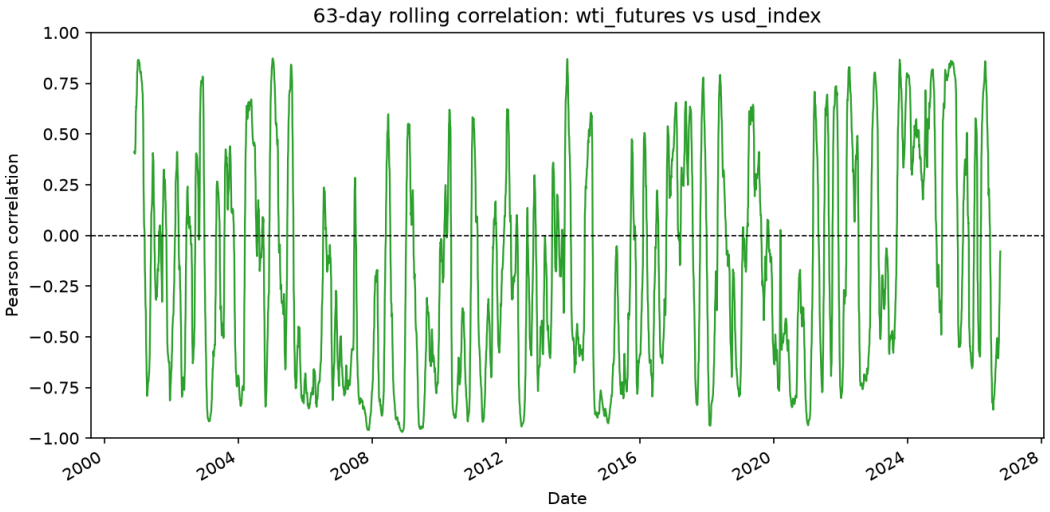


Figure 5: 63-day rolling correlation, WTI futures vs. the US dollar index.

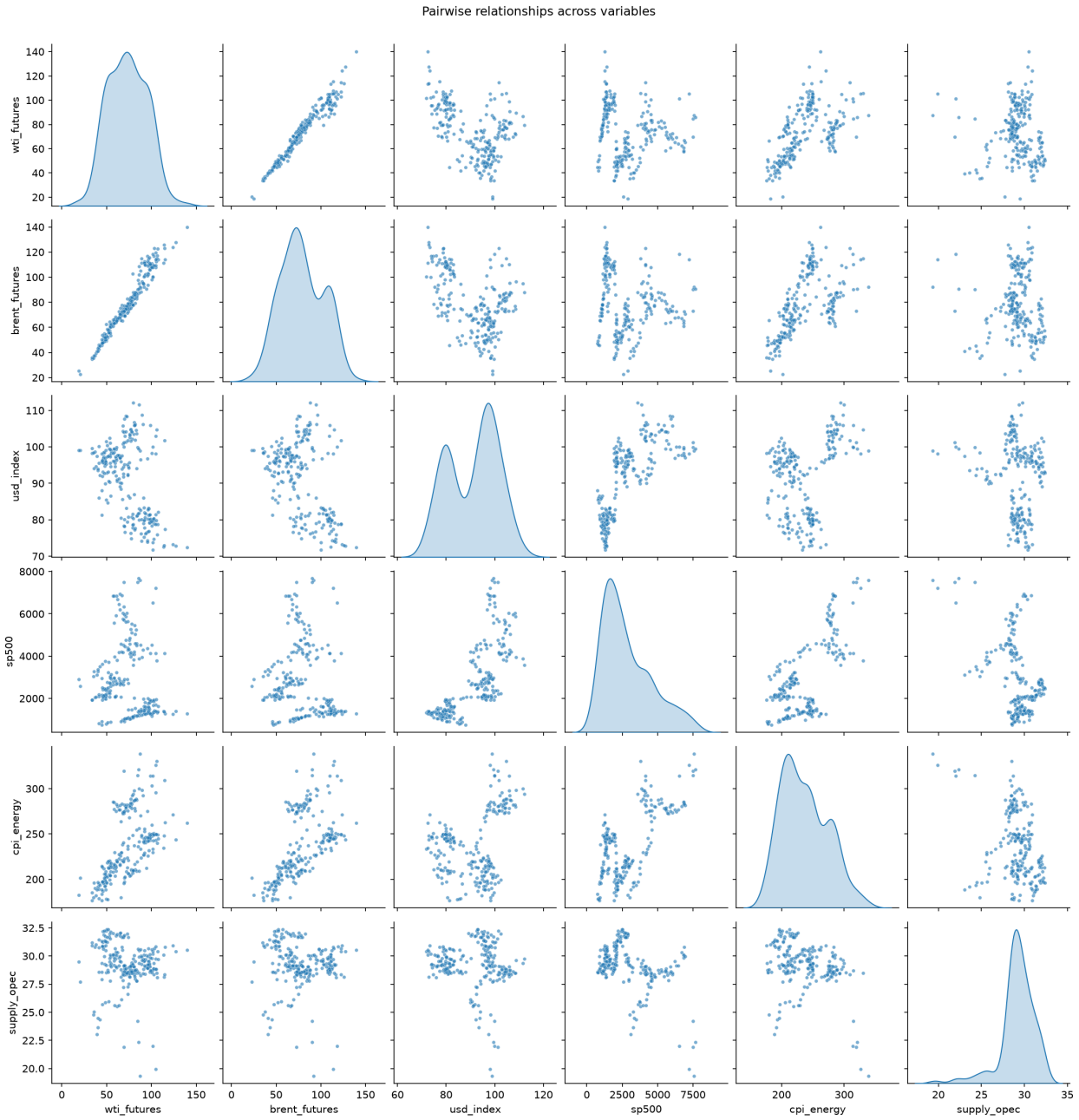


Figure 6: Scatter matrix of a curated headline subset (both crude benchmarks, the US dollar index, the S&P 500, the energy CPI index, and OPEC supply), resampled to month-end.

4 Stylized Facts

Here we use statistics computed on daily log returns (see Table 2) for the two crude benchmarks and the above figures for our analysis.

(a) The level plots (Figure 2) show oil behaving less like a smoothly-trending equity index

Table 2: Stylized-facts statistics on daily log returns.

	WTI	Brent
Observations	6,557	4,775
Annualised mean return	3.87%	1.39%
Annualised volatility	42.6%	38.7%
Skewness	−0.51	−0.77
Excess kurtosis	18.74	11.63
Jarque-Bera statistic	96,264	27,389
Jarque-Bera p-value	< 0.001	< 0.001
ACF of returns, lag 1	−0.004	−0.018
ACF of squared returns, lag 1	0.46	0.20
ACF of squared returns, lag 5	0.25	0.12

and more like a regime-switching series: multi-year bull/bear swings (the 2000-08 rally, the 2008-09 collapse, the 2014-16 collapse, the 2020 COVID crash, and the 2022 spike) punctuated by sharp, short-lived shocks. The rolling-volatility plot shows that volatility itself is clustered and regime-dependent, with calm multi-year stretches interrupted by violent spikes exceeding 2.5× the typical annualised level in April 2020. That episode is itself a uniquely oil-specific phenomenon: as discussed in Section 2.2, WTI's near-month contract briefly traded at a negative price on 20 April 2020 — essentially impossible for an equity or FX price, but a direct consequence of physical-delivery settlement when storage capacity is exhausted. Our general-purpose cleaning pipeline treats that single observation as bad data and imputes it away, which is a transparent, documented trade-off rather than a silent one, and the event itself remains a genuine, worth-noting stylized fact about oil specifically.

(b) WTI returns show mild negative skewness (−0.51, i.e. large down-moves are somewhat more extreme than large up-moves) and strongly leptokurtic, fat-tailed behaviour (excess kurtosis of 18.74 against 0 for a Normal distribution); Brent shows the same pattern (skewness −0.77, excess kurtosis 11.63). The Jarque-Bera test rejects normality overwhelmingly for both series ($p < 0.001$), consistent with the pronounced bowing away from the theoretical line visible in the Q-Q plot (Figure 1) beyond roughly ±2 standard deviations.

(c) Essentially none at the level of raw returns: the lag-1 autocorrelation is −0.004 (WTI) and −0.018 (Brent), and the ACF/PACF plot (Figure 3) shows no economically significant spikes beyond lag 0 — consistent with (weak-form) market efficiency, i.e. the sign of tomorrow's return is not linearly predictable from today's. The squared returns, however, tell a very different story: lag-1 autocorrelation of squared returns is 0.46 (WTI) and 0.20 (Brent), still 0.25/0.12 at lag 5 — a strong, slowly-decaying autocorrelation. This is the textbook signature of volatility clustering: returns themselves are unpredictable, but the magnitude of returns is highly persistent, motivating a regime-switching (Hidden Markov Model) approach.

(d)

- A negative, but highly unstable, correlation with the US dollar index: the full-sample level correlation is -0.57 (WTI) / -0.46 (Brent) (Figure 4), yet the 63-day rolling correlation (Figure 5) oscillates repeatedly between roughly -1 and $+1$ across the whole sample with no single stable regime — the textbook "stronger dollar, cheaper oil" relationship holds on average but is far too unstable to use as a reliable short-horizon hedge on its own.
- WTI and Brent correlate at 0.98 in levels and track each other almost one-for-one, as expected for two prices of the same underlying global commodity linked by arbitrage.
- Oil correlates positively, but only moderately, with broad inflation/risk-sentiment variables — the energy CPI index (0.77 WTI, 0.58 Brent) more so than the S&P 500 (0.19 WTI, -0.11 Brent) — i.e. energy prices are a meaningful component of realised inflation, not merely a passive responder to broad equity risk sentiment.
- Geopolitical/supply variables correlate only weakly and inconsistently with price (e.g. OPEC supply: 0.27 WTI / -0.08 Brent; non-OPEC supply: 0.29 WTI / -0.24 Brent) — far weaker than the financial variables above — consistent with our central thesis (Section 1) that oil-price dynamics are better captured by a non-linear, graphical/probabilistic model of many interacting physical and financial drivers than by simple pairwise linear correlation.

5 Probabilistic Graphical Models: Belief Networks and Parameter Learning

5.1 Belief Networks

A belief network (also called a Bayesian network) represents the joint probability distribution of a set of random variables $X_1, \ldots, X_m$ as a directed acyclic graph (DAG) $G = \langle V, E \rangle$, where each node $v \in V$ corresponds to one variable X_v, and each directed edge encodes a direct probabilistic dependency. The defining property of this representation is that the joint density factorises along the graph structure:

$$p(x_1, \ldots, x_m) = \prod_{v=1}^{m} p\big(x_v \mid \mathrm{pa}(x_v)\big),$$

where $\mathrm{pa}(x_v)$ denotes the values taken by the parents of v in G (and $p(x_v \mid \mathrm{pa}(x_v)) = p(x_v)$ for a node with no parents). This factorisation is what makes belief networks tractable: instead of specifying one joint distribution over all m variables jointly, the modeller only needs to specify m local conditional distributions, one per node, each depending only on that node's (typically small) parent set (Pearl; Koller and Friedman).

A standard illustrative example is the "Rain–Sprinkler–Wet Grass" network: whether the grass is wet depends on whether it rained and whether the sprinkler was on, and both of those, in turn, have their own (unconditional) probabilities of occurring. Once the network and its conditional probability tables are specified, exact inference lets us answer queries in the reverse causal direction — e.g. given that we observe the grass is wet, what is the probability that it rained? — via Bayes' theorem and marginalisation over the remaining variables. The same machinery applies, at much larger scale, to our 24-variable oil-price panel: observing a change in, say, OPEC spare capacity and the US dollar index lets the network update its belief about the probability distribution of the oil-price node, without needing a separate model for every possible combination of observed evidence.

5.2 Learning a Belief Network

Given a dataset of observations but no pre-specified network, learning a belief network splits into two distinct sub-problems:

Parameter learning Given a fixed DAG G that already captures the dependency structure between variables, estimate the conditional probability distribution (CPD) at each node, i.e. estimate the parameters $\theta_{v|\mathrm{pa}(v)}$ of $p_{v|\mathrm{pa}(v)}(x_v; \theta_{v|\mathrm{pa}(v)})$.

Structure learning Given only the raw data, estimate the DAG G itself, i.e. decide which arcs are present, without yet committing to any parameter values. This is covered in Section 6.

This section focuses on parameter learning. One of the principal practical advantages of the graphical representation is that it decomposes one high-dimensional estimation problem (the full joint distribution over all m variables) into m independent, low-dimensional estimation problems, one per node, each conditioned only on that node's parents. This is only computationally cheap if the graph is reasonably sparse: the number of free parameters at a node grows with the number (and cardinality) of its parents, so there is a direct computational incentive to prefer sparser structures — a consideration that resurfaces in Section 6 when scoring candidate structures.

Two standard estimators are used for the CPD parameters at each node:

Maximum Likelihood Estimation (MLE). The CPD at each node is filled in using the observed relative frequencies of the variable's states (for discrete data), choosing the parameter values that maximise $P(\text{data} \mid \text{model})$. This is simple and consistent, but can behave poorly when some parent-value combinations are rarely or never observed in the sample (a combination with zero observed occurrences receives zero probability, which is often too strong a conclusion to draw from a finite sample).

Bayesian Estimation. Rather than starting from an uninformative (empirical-frequency) prior, Bayesian parameter estimation begins from an explicit prior CPD expressing the modeller's belief about the parameters before observing the data, and updates it with the observed data via Bayes' rule. This naturally regularises the rare/unseen parent-combination problem that pure MLE struggles with, at the cost of requiring the modeller to choose a (possibly subjective) prior.

5.3 Relevance to this Project

In our oil-price panel, the "variables" that ultimately enter the belief network are not the raw, continuous series in Table 1, but discretised regime labels produced by the Hidden Markov Model stage (e.g. "WTI: bull/stagnant/bear", "USD index: strengthening/stable/weakening"). Once each series has been reduced to a small number of discrete regimes, parameter learning reduces to exactly the discrete-CPD estimation problem described above, and can be carried out directly with standard tools (e.g. `pgmpy`'s maximum-likelihood or Bayesian estimators (Ankan and Panda)) once the network structure itself has been learned, as discussed next.

6 Structure Learning: Constraint-Based and Score-Based Methods

Structure learning is the harder of the two belief-network learning problems introduced in Section 5: given only a dataset, find the DAG G that best represents the dependencies between variables, without assuming the structure in advance. The search space is enormous — for n variables there are $n(n-1)/2$ possible undirected edges and therefore up to $2^{n(n-1)/2}$ candidate structures — so an exhaustive search is computationally infeasible beyond a handful of variables, and the literature instead relies on two broad families of approximate algorithm: constraint-based and score-based learning.

6.1 Constraint-Based Learning

Constraint-based algorithms use statistical conditional-independence tests (e.g. Pearson's χ^2 test for categorical data, or Fisher's Z-test for Gaussian data) to directly determine which arcs belong in the graph, rather than optimising a global score. Every constraint-based algorithm shares the same three-phase structure, illustrated by the Inferred Causality algorithm described in Section 8:

- learn each node's Markov blanket to prune the candidate structure early;
- use the reduced candidate set to identify which undirected arcs (the "skeleton") are present
- orient the remaining undirected arcs into a final, partially-directed acyclic graph.

The appeal of this family is that it directly tests the conditional-independence relationships that a DAG is meant to represent, and the Markov-blanket pruning step (Section 7) gives a substantial computational saving over testing every possible subset of variables. Its main weakness is sensitivity to the statistical independence tests themselves: with a finite, noisy sample (as in our 24-variable panel, where some EIA series only begin in 2003), individual test errors can propagate into structural errors that are hard to correct downstream.

6.2 Score-Based Learning

Score-based algorithms instead assign a numerical score $\mathrm{Score}(G, D)$ to every candidate structure G, quantifying how well G represents the observed dataset D, and search for the structure that maximises it. Writing the score as the posterior probability of the graph given the data,

$$\mathrm{Score}(G, D) = \Pr[G \mid D] \;\propto\; \Pr[D \mid G]\,\Pr[G],$$

a common and well-behaved choice is the **Bayesian Information Criterion (BIC)**,

$$\mathrm{BIC}(G, D) = \sum_{v \in V} \log \hat{p}_{v|\mathrm{pa}(v)}\big(X_v \mid \mathrm{pa}(v)\big) - \tfrac{1}{2} k_{v|\mathrm{pa}(v)} \log n,$$

where $k_{v|\mathrm{pa}(v)}$ is the number of free parameters at node v given its parents. The first term rewards structures that fit the data well; the second penalises structures with more free parameters, which both discourages overfitting and offsets BIC's bias towards overly simple structures as the sample size n grows.

Because an exhaustive search over all candidate DAGs remains infeasible even with a scoring function in hand, score-based learning is typically paired with a local search heuristic. The most widely used is **Hill Climbing**: starting from an initial structure (empty, full, random, or seeded with expert knowledge), the algorithm repeatedly tries every single-edge modification available at the current structure — adding an edge, removing an edge, or reversing an edge's direction — and moves to whichever neighbouring structure most improves the score. The procedure halts when no single-edge change improves the score further. Hill Climbing is greedy and therefore has no guarantee of finding the globally optimal structure; it can settle at a local maximum depending on its starting point. Standard mitigations include a tabu list of recently-visited structures (to discourage immediately undoing the last move) and random restarts from a different starting structure.

6.3 Trade-offs and Relevance

Constraint-based and score-based learning make different trade-offs: the former directly tests the conditional-independence structure the DAG is supposed to represent, at the cost of sensitivity to individual test errors; the latter avoids relying on any single statistical test by instead optimising a global, sample-size-aware score, at the cost of only finding a local optimum via a greedy search. For this project, we use `pgmpy`'s `HillClimbSearch` with a BIC score as the primary score-based structure-learning tool, since it scales comfortably to our panel size, requires no manual tuning of independence-test thresholds, and its score explicitly trades off fit against the number of free parameters — a desirable property given how many of our 24 series are themselves strongly collinear (Figure 4), which would otherwise encourage a constraint-based test to retain spurious edges. We separately implement the constraint-based Inferred Causality algorithm by hand in Section 8, both as a pedagogical exercise and as a cross-check on the Hill-Climbing result.

7 Markov Chains, Hidden Markov Models, and the Markov Blanket

7.1 Markov Chains

A Markov Chain is a graphical abstraction of a stochastic process that transitions between a finite number of states, used to model systems whose future evolution depends only on their present state and not on the full history that led to it. This memorylessness property is formalised as the **Markov property**: for a discrete-time process $\{X_t, t \geq 0\}$ taking values in a countable state space,

$$\Pr[X_{t+1} = j \mid X_t = i, X_{t-1} = k, \ldots, X_0 = m] = \Pr[X_{t+1} = j \mid X_t = i] = p_{ij}, \qquad \forall t \geq 0.$$

A finite-state Markov chain with N states can be represented either as a graph (a node per state, a directed weighted arc for every non-zero transition probability p_{ij}) or, equivalently, as an $N \times N$ row-stochastic transition matrix $[M] = \{p_{ij}\}$ with $\sum_j p_{ij} = 1$ for every i. This duality is useful in practice: the graphical view makes the qualitative structure of the process easy to inspect, while the matrix view makes it straightforward to apply standard linear-algebra machinery (e.g. computing the stationary distribution as the matrix's leading eigenvector).

7.2 Hidden Markov Models

A **Hidden Markov Model (HMM)** extends a Markov chain to the case where the states themselves are not directly observed. Formally, an HMM is a 5-tuple (Q, Σ, Π, A, B), where $Q = \{q_1, \ldots, q_N\}$ is a finite set of hidden states, $\Sigma = \{s_1, \ldots, s_M\}$ is a finite set of observable symbols, $\Pi = \{\pi_i\}$ is the initial-state probability vector, $A = \{a_{ij}\}$ is the hidden-state transition matrix, and $B = \{b_i(v_k)\}$ is the emission-probability matrix giving the probability of observing symbol v_k while in hidden state i. The transition between hidden states is itself assumed to follow a first-order Markov chain as defined above; only the emissions generated by each hidden state are observed directly.

Three canonical problems are associated with an HMM $\lambda = (\Pi, A, B)$, each with a classical dynamic-programming solution:

Evaluation Compute the probability $p(O \mid \lambda)$ of an observed emission sequence $O = o_1 o_2 \cdots o_T$ under the model — solved by the **Forward** and **Backward** algorithms.

Decoding Find the most likely hidden-state path $q^* = \arg\max_q p(q, O \mid \lambda)$ that produced an observed sequence — solved by the **Viterbi** algorithm.

Training Find the parameters $\lambda^* = \arg\max_\lambda p(O \mid \lambda)$ that best explain an observed sequence

— solved by the **Baum-Welch** algorithm (an instance of Expectation-Maximisation that repeatedly calls the Forward and Backward algorithms to re-estimate Π, A, and B) (Rabiner and Juang).

In this project, the HMM's hidden states correspond to latent market regimes (e.g. bull/bear/stagnant), while the emissions are the observed values (or discretised bins) of each financial/macro/geopolitical series.

7.3 The Markov Blanket

The **Markov blanket** of a node v in a DAG is defined as the union of v's parents, v's children, and the other parents of v's children (its children's co-parents) (Pearl). Its defining property is that it is a minimal sufficient summary of everything else in the network relevant to v: every other node in the graph is conditionally independent of v once the Markov blanket of v is known, i.e.

$$X_v \perp\!\!\!\perp X_u \mid B(X_v) \qquad \text{for every } u \notin B(X_v) \cup \{v\}.$$

This property has two direct, practical consequences for this project. First, it gives risk management a precise, minimal answer to "what do I need to monitor to explain the oil-price node?": once the price node's Markov blanket is known, every other variable in the 24-series panel becomes redundant for predicting it. Second, and more immediately, it is the computational engine behind the constraint-based Inferred Causality structure-learning algorithm developed next (Section 8): learning each node's Markov blanket first lets the algorithm discard the overwhelming majority of candidate conditioning sets before testing for undirected arcs, which is what makes constraint-based learning tractable at all on a panel of this size (Pellet and Elisseeff).

8 The Inferred Causality Algorithm

As introduced in Section 6, every constraint-based structure-learning algorithm shares a common three-phase skeleton: learn each node's Markov blanket, use it to identify the undirected skeleton of the graph, and finally orient the remaining arcs. **Inferred Causality** is the canonical algorithm of this family (Pellet and Elisseeff; Alvi). Below is our own pseudocode rendering of the algorithm, written from our understanding of its three phases rather than reproduced from any single source, followed by a short walk-through of why each phase is structured the way it is.

Algorithm 1 Inferred Causality (our rendering, after Pellet & Elisseeff (2008) and Alvi (2018))

Require: Dataset D over variables $X_1, \dots, X_m$
Ensure: A partially directed acyclic graph G over $X_1, \dots, X_m$

Phase 1 — Learn Markov blankets

for each variable X_i **do**
    Estimate its Markov blanket $B(X_i)$ from D (e.g. via an independence-test-based feature-selection search).
for each pair (X_i, X_j) **do**
    if $X_i \in B(X_j)$ **xor** $X_j \in B(X_i)$ **then**
        Discard the asymmetric membership as a false positive, so that $B(\cdot)$ is made symmetric: $X_i \in B(X_j) \Leftrightarrow X_j \in B(X_i)$.

Phase 2 — Learn the undirected skeleton

for each pair (X_i, X_j), $i \neq j$ **do**
    Search for a conditioning set $S_{ij} \subseteq V \setminus \{X_i, X_j\}$ such that $X_i \perp\!\!\!\perp X_j \mid S_{ij}$. **If** $B(X_i)$ and $B(X_j)$ are already known from Phase 1, restrict the search to subsets of the smaller of $B(X_i) \setminus \{X_j\}$ and $B(X_j) \setminus \{X_i\}$ (the Markov-blanket pruning that makes this phase tractable).
    if no such S_{ij} is found **then**
        Place an undirected arc $X_i - X_j$ in the skeleton.
Re-symmetrise the resulting neighbour sets $N(X_i)$ exactly as in Phase 1, discarding asymmetric neighbour relations.

Phase 3 — Orient the arcs

for each pair of *non-adjacent* variables X_i, X_j with a common neighbour $X_k \notin S_{ij}$ **do**
    Orient both arcs away from X_k: set $X_i \rightarrow X_k \leftarrow X_j$ (a *v-structure* / collider).
repeat
    if $X_i - X_j$ is still undirected and a strictly directed path already exists from X_i to X_j **then**
        Orient $X_i \rightarrow X_j$ (orienting it the other way would create a directed cycle).
    if X_i, X_j are non-adjacent, $X_i \rightarrow X_k$, and $X_k - X_j$ is still undirected **then**
        Orient $X_k \rightarrow X_j$ (otherwise $X_i \rightarrow X_k \leftarrow X_j$ would become an unintended second v-structure at X_k).
until no further arc can be oriented
return the resulting partially directed acyclic graph G.

8.1 Discussion

The structure of Algorithm 1 is best understood as successive narrowing of the search space. Phase 1 uses the Markov blanket's defining property (Section 7) purely as a pruning device: instead of testing every one of the 2^{m-2} possible conditioning sets for every pair of variables, Phase 2 only has to search within each node's own (already small) Markov blanket. The symmetry checks in both Phase 1 and Phase 2 exist because the independence tests used to estimate $B(X_i)$ are statistical, not exact, so sample noise can produce $X_i \in B(X_j)$ without the reverse holding; treating such asymmetries as false positives trades a small amount of recall for a large improvement in precision. Phase 3 then exploits a different structural fact: once the skeleton is fixed, only the v-structures (colliders) are identifiable from conditional-independence information alone, since they are the only local configuration where the Markov-blanket's conditional independencies pin down a unique orientation; every other arc is oriented only indirectly, by propagating the acyclicity constraint from the already-oriented v-structures.

For our oil-price panel, we treat this algorithm as a cross-check on the score-based Hill-Climbing search (Section 6) that we will use as our primary structure-learning tool in GWP2: because Inferred Causality and Hill-Climbing make different assumptions (local conditional-independence tests vs. a global, parameter-penalised score), arcs recovered by both methods are a stronger signal of a genuine dependency in the underlying oil market than an arc recovered by either method alone.

Works Cited

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